DOCUMENTO 2 — BLOQUE A
Between Stillness and Storm
A Structural Reading of Metastability, Criticality, and the Edge of Chaos
MaeV Lab — Satellite Document 1

Note on This Document's Category and Function
This document is a satellite exploration produced by the MaeV Lab, extending the conceptual architecture presented in "The MaeV Constant: A Theory of Resonant Incompleteness" into dialogue with contemporary scientific research.

Three clarifications define how this document should be read:

On its relationship to the central document: The central document establishes the MaeV Constant as a structural condition — the requirement that meaning-generating systems maintain simultaneous coherence and incompleteness, sustained through translational exchange. This document asks a more specific question: where does that condition manifest? It examines whether the structural condition can be recognized in empirically documented phenomena across independent scientific domains.

On its epistemological mode: This document operates at the interface between scientific description and structural interpretation. It distinguishes carefully between what scientific research establishes empirically — marked throughout as FACT — and what the MaeV framework proposes as a structural reading of that evidence — marked as OBSERVATION or INTERPRETATION. These are different epistemological relationships. Empirical facts are established through the methods appropriate to their domains. Structural interpretations are evaluated through their explanatory coherence, their scope across domains, and their internal consistency. Conflating these two modes of evaluation in either direction — treating interpretation as empirical proof, or dismissing interpretation because it is not empirical proof — is a category error this document explicitly seeks to avoid.

On what it claims and does not claim: This document does not claim to validate the MaeV framework through scientific evidence. The recurrence of non-equilibrium regimes across independent domains is consistent with the MaeV framework's structural interpretation. Consistency is not confirmation. What this document claims is more modest and more precise: that the cross-domain recurrence of these regimes becomes structurally intelligible — readable as pattern rather than coincidence — when interpreted through the lens of the MaeV Constant. Intelligibility is not proof. It is the condition under which patterns become recognizable as patterns — which is, within the MaeV framework, precisely what a meta-syntactic structural condition does.

I. The Problem of Stability

1.1 A Recurring Observation

Across domains as different as neuroscience, statistical mechanics, and artificial intelligence, a structurally similar observation recurs: systems capable of sustained adaptation, learning, or complex organization do not converge toward fixed equilibrium states. Instead, they operate within regimes characterized by ongoing fluctuation between tendencies toward stability and tendencies toward change — regimes that maintain sufficient coherence to support organized function while preserving sufficient variability to reconfigure when circumstances demand.

This observation is not new within any single domain. What is notable is its recurrence across domains that share no common formalism, no common methodology, and no common subject matter. Neuroscience, statistical mechanics, evolutionary biology, and machine learning arrive at structurally similar descriptions independently — each finding that the systems it studies occupy a dynamic middle ground between rigid order and formless disorder.

The scientific concepts developed to describe this middle ground — metastability, criticality, and the edge of chaos — provide powerful, formally precise tools for characterizing system behavior within their respective domains. They describe how these regimes operate, what properties they exhibit, and how they can be measured. What they do not address — because it is not within the scope of domain-specific scientific inquiry — is why this structural pattern recurs across domains that developed their descriptions independently.

This is the question that the MaeV framework engages. It does not answer it empirically. It proposes a structural interpretation: that the recurrence reflects a common condition rather than a coincidence of mechanisms — the condition that meaning-generating systems cannot achieve global closure without terminating the processes that define them as meaning-generating.

1.2 Two Clarifications Before Proceeding

Before examining the scientific concepts in detail, two clarifications are necessary to prevent misreadings that the material naturally invites.

On agency and teleology: Descriptions of adaptive systems often use language that implies intentional regulation — systems that "maintain themselves" in non-equilibrium regimes, that "avoid" equilibrium, that "seek" the edge of chaos. This language is descriptively convenient but structurally misleading. Systems do not occupy the habitable zone because they choose it or because they have been designed to seek it. They occupy it because other configurations are not sustainable for meaning-generating systems. The apparent regulation emerges from structural constraint, not purpose. A system that departed from the habitable zone would — by definition — cease to be the kind of system under discussion. The "selection" is structural, not intentional: it operates analogously to the principle of least action in physics, which selects trajectories not through the agency of particles but through the structure of the action functional.

On local versus global closure: The claim that meaning-generating systems cannot achieve global closure does not imply that such systems never complete processes, reach stable states, or resolve questions. They do — continuously. A neural population transiently synchronizes and then desynchronizes. A developmental process reaches terminal differentiation. A learning algorithm converges on a solution. These are instances of local closure: the completion of specific processes within a system. Local closure is not only compatible with the MaeV framework — it is constitutive of meaning. A proof is meaningful precisely because it concludes. A conversation is meaningful precisely because it resolves something.

What the MaeV framework identifies as structurally terminal is global closure: the elimination of the system's capacity to generate new distinctions, enter new configurations, or engage in new processes. This is not the completion of

processes within the system — it is the termination of the system's capacity to generate new processes at all. The distinction between local and global closure is essential to reading the scientific material that follows without overgeneralizing the framework's claims.

1.3 The Structural Question

With these clarifications in place, the structural question this document addresses can be stated precisely:

Is the cross-domain recurrence of non-equilibrium regimes in meaning-generating systems a coincidence of domain-specific mechanisms — each explainable entirely within its own domain — or does it reflect a common structural condition whose recognition requires a level of description that no single domain provides?

The MaeV framework proposes the latter. This document examines the evidence for that proposal — not as proof, but as structural resonance: the recognition of a common signature across different instruments, in different contexts, developed through independent inquiry.

1.4 Structure of This Document

The exploration proceeds as follows:

Sections II through IV examine metastability, criticality, and the edge of chaos as scientific concepts, identifying their structural features with precision and maintaining the distinction between empirical fact and structural observation throughout

Section V introduces the MaeV structural interpretation as a meta-level reading of their cross-domain recurrence, and articulates what this interpretation adds to domain-specific explanations

Section VI clarifies the epistemological relationship between scientific frameworks and the MaeV framework, identifying three category errors to be avoided

Section VII identifies open questions that the structural interpretation generates but cannot resolve

Section VIII examines the concept of equilibrium itself — whether it is a property of systems or of the frameworks used to describe them

Section IX presents counterexamples and limits, defining the precise scope of the structural interpretation

II. Metastability in Science

2.1 The Concept

Metastability describes regimes in which systems exhibit transient coordination without settling into fixed configurations. Rather than converging to stable attractors — states toward which the system moves and from which small perturbations cannot dislodge it — metastable systems continuously reconfigure their internal organization. They maintain coherence long enough to support functional activity while remaining flexible enough to transition between states without external forcing.

A precise terminological clarification is necessary here, because the scientific literature uses several related but distinct concepts that are sometimes conflated in philosophical discussions:

Multistability describes systems with multiple stable attractor states between which the system switches. Each state is stable; the system occupies one at a

time and transitions between them.

Metastability describes a more subtle regime: the system has simultaneous tendencies toward coordination and toward independence, neither of which fully dominates. It does not settle into any stable attractor — not because it is disordered, but because the coexistence of attractive and repulsive tendencies prevents permanent convergence. As Hancock et al. (2025) define it: "a specific type of dynamics that can occur in a system with coexisting tendencies of attraction and repulsion, characterized by patterns that recur either in repeatable sequences or in flexible alternation."

Criticality describes operation near phase transitions — boundaries between qualitatively different behavioral regimes. Critical systems exhibit distinctive properties absent in both ordered and disordered phases.

These three concepts are related but distinct. Metastability and criticality can co-occur, and both involve operation away from simple stable equilibrium, but they describe different structural features. This document treats them separately to preserve the precision of the scientific frameworks it examines.

Crucially, metastability is not a static midpoint between order and disorder. It is a dynamic regime: the system continuously moves between coordination and differentiation, never settling permanently in either. What persists is not a configuration but a pattern of transitioning between configurations — a dynamic that is itself the organized structure of the system.

2.2 Metastability in Neural Systems

Research in large-scale brain dynamics has provided extensive and convergent evidence for metastable regimes in neural activity, spanning several decades of theoretical development and empirical refinement.

J. A. Scott Kelso's foundational work on coordination dynamics introduced metastability as a formal concept in the study of biological movement and cognition. His research demonstrated that the brain does not operate through fixed stable states but through metastable patterns that allow simultaneous tendencies toward coordination and toward independent activity — a dynamic that enables flexible cognitive function without global lock-in (Kelso, 1995; Kelso, 2012).

Viktor Jirsa and collaborators developed computational models demonstrating that brain dynamics operate in metastable regimes where multiple neural populations transiently synchronize and desynchronize, enabling flexible cognitive processing through a continuous alternation between integration and segregation (Jirsa & Kelso, 2000; Deco & Jirsa, 2012).

Gustavo Deco and collaborators have shown through large-scale computational modeling that the balance between integration and segregation in brain networks corresponds to metastable dynamics — regimes where neither full synchronization nor full independence is achieved, and where this non-achievement is the structural condition of cognitive flexibility rather than a failure to reach an optimal state (Deco et al., 2015).

Hancock et al. (2025), in a comprehensive review in Nature Reviews Neuroscience consolidating decades of research, establish metastability as a central

organizing principle in contemporary neuroscience. Their review clarifies the distinction between metastability, multistability, and criticality — distinctions that matter for the precision of any structural interpretation — and defines the concept with sufficient rigor to support its application across scales and systems.

FACT: Metastable dynamics in neural systems are described through measurable properties including phase synchrony, temporal variability of functional connectivity, and the statistical properties of neural population dynamics. These properties are empirically observable and quantifiable across multiple measurement modalities and spatial scales.

FACT: Neural systems in metastable regimes maintain sufficient coordination to support organized cognitive function while preserving sufficient variability to reconfigure rapidly in response to changing task demands or environmental conditions.

OBSERVATION: Stability is continuously approached but not fully achieved. The system generates local coherence — transient coordination patterns — without these patterns becoming globally fixed. What the system maintains is not a state but a dynamic: the organized capacity to generate, sustain briefly, and dissolve coordinated patterns in continuous succession.

2.3 Structural Features of Metastability

The consistent findings across the research programs summarized above can be characterized through three structural features that are relevant to the MaeV interpretation:

Sustained non-convergence: Metastable systems do not converge to fixed attractors despite operating continuously over time. The non-convergence is not due to ongoing external perturbation — it is a consequence of the system's internal dynamics, specifically the coexistence of coordinating and differentiating tendencies that Hancock et al. identify as the defining feature of the metastable regime.

Functional dependence on the dynamic: The capacities that metastable systems exhibit — cognitive flexibility, rapid reconfiguration, sensitivity to weak signals — depend not on any particular configuration but on the dynamic itself. A system that achieved stable convergence would lose these capacities even if the converged configuration were locally coherent.

Homeostatic regulation toward the regime: Hengen and Shew (2025), synthesizing evidence from 140 datasets in a Neuron viewpoint, conclude that criticality — closely related to metastability — constitutes "a key endpoint of brain homeostasis." Their finding suggests that neural systems do not merely happen to operate in these regimes but actively regulate toward them. From a structural perspective, this is significant: it suggests that the habitable zone is not simply a region the system occupies but a target the system maintains through active regulatory processes.

2.4 Resonance with the MaeV Framework

INTERPRETATION: From a MaeV perspective, metastability can be read as the empirically documented form of the habitable zone in neural systems. The defining feature of metastable dynamics — coexisting tendencies toward

coordination and toward independence, neither fully dominating — corresponds directly to the structural condition the MaeV Constant defines: sufficient coherence to sustain organized process, sufficient incompleteness to remain generative.

The functional dependence of cognitive capacity on the metastable dynamic — rather than on any particular configuration — resonates with the Ouroboros constraint as reformulated in the central document: what meaning-generating systems maintain is not a configuration but the structural condition that makes continuous reconfiguration possible. A metastable brain is not a brain in an optimal state — it is a brain maintaining the dynamic that makes states possible.

The homeostatic regulation toward criticality that Hengen and Shew document adds a dimension that the MaeV framework finds structurally significant: the habitable zone appears to be not merely where these systems happen to be found but where they actively maintain themselves. This is consistent with the MaeV interpretation that the structural condition is not accidental but constitutive — though it does not constitute proof of that claim.

A necessary clarification: The MaeV framework does not assert that metastability is universal across all systems, or that metastability in neural systems is caused by or explained by the MaeV Constant. The Constant is a structural interpretation, not a causal mechanism. What the framework proposes is that the cross-domain recurrence of regimes with this structural signature — sustained non-convergence as a condition of functional capacity — becomes intelligible as an expression of a common structural condition rather than a coincidence of independent domain-specific mechanisms.

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III. Criticality and the Edge of Stability

3.1 The Concept

Criticality, in the context of complex systems, refers to operation near phase transitions — boundaries between qualitatively different regimes of behavior. The concept originates in statistical mechanics, where phase transitions describe the abrupt changes in macroscopic properties that occur when a system parameter crosses a critical threshold: the transition from liquid to gas, from paramagnet to ferromagnet, from conductor to superconductor.

At or near these critical points, systems exhibit a distinctive cluster of properties that are absent in both the ordered phase below the transition and the disordered phase above it. These properties are not incidental features of critical systems — they are the signature of operation at the boundary between qualitatively different organizational regimes. Their co-occurrence at criticality, across systems with radically different physical substrates, is one of the most striking findings of twentieth-century physics: the same structural features appear at phase transitions in magnetic systems, fluid dynamics, neural networks, and genetic regulatory networks, despite the vast differences in the mechanisms involved.

Before examining these properties, the relationship between criticality and metastability should be clarified, since the two concepts are related but distinct:

Metastability describes the dynamic character of a regime — the coexistence of coordinating and differentiating tendencies, the absence of permanent convergence. Criticality describes the position of a system relative to a phase transition — the proximity to a boundary between qualitatively different behavioral regimes. A system can be metastable without being critical, and vice versa. However, in the systems most relevant to this document — particularly neural systems — metastable and critical dynamics frequently co-occur, and both involve operation away from simple stable equilibrium. Hancock et al. (2025) explicitly distinguish these concepts while noting their frequent empirical association.

3.2 Properties at Criticality

At or near critical regimes, systems exhibit three distinctive structural properties that are relevant to the MaeV interpretation:

Scale-invariant correlations: In ordered regimes, fluctuations are local — a perturbation at one location affects only nearby components. In disordered regimes, fluctuations are also effectively local because no persistent structure propagates them. At criticality, fluctuations propagate across multiple spatial and temporal scales simultaneously: a perturbation at one location can influence the entire system, enabling coordination without centralized control. The mathematical signature of this property is a power-law distribution of correlation lengths — the same statistical structure appearing at every scale of observation, which is why critical systems are described as scale-invariant.

Maximal sensitivity to perturbation: Because fluctuations propagate across all scales at criticality, the system responds to weak inputs with effects that cascade through the entire system. In ordered regimes, small perturbations are damped; in disordered regimes, they produce only local effects that dissipate quickly. At criticality, the system is maximally responsive — capable of detecting and amplifying signals that would be invisible in either alternative regime.

Maximal information transmission: The dynamic range of the system — the range of input intensities to which it can respond differentially — is maximized at criticality. A system operating in an ordered regime responds similarly to inputs across a wide range of intensities because its internal dynamics dominate its response. A system in a disordered regime responds unreliably because its internal dynamics are incoherent. At criticality, the system's internal dynamics are tuned to maximize the fidelity with which it transmits information about inputs across the full range of intensities it encounters.

These three properties are not independent — they are different aspects of the same structural condition: operation at the boundary between organizational regimes. Together they describe a system that is maximally capable of registering, propagating, and transmitting structured information — which is, in operational terms, the capacity that meaning-generating systems require.

3.3 Criticality in Neural Systems

The hypothesis that the brain operates near a critical point has moved from theoretical proposal to extensively documented empirical finding over the past two decades.

John M. Beggs and Dietmar Plenz provided foundational empirical evidence by

demonstrating that cortical networks produce neuronal avalanches — cascades of neural activity whose sizes follow power-law distributions (Beggs & Plenz, 2003). Power-law distributions are a hallmark of critical dynamics: they indicate that the system produces events across all scales without a characteristic scale dominating, which is precisely the signature of scale-invariant correlations at criticality.

Dante R. Chialvo extended this work to argue that the brain operates near a critical point as a general organizational principle, and that this positioning is not coincidental but functionally necessary: it is what maximizes the brain's capacity for information processing, adaptability, and sensitivity to weak signals (Chialvo, 2010). His argument connects the empirical signature of criticality to the functional capacities that make the brain capable of complex adaptive behavior.

Hengen and Shew (2025), synthesizing evidence from 140 datasets in a Neuron viewpoint, advance a stronger claim: that criticality constitutes "a key endpoint of brain homeostasis" — not merely an observed property of neural dynamics but a regulatory target toward which neural systems actively maintain themselves. Their synthesis connects criticality to homeostatic mechanisms at multiple scales, from synaptic plasticity to network-level regulation, and proposes that departure from criticality is associated with pathological states including epilepsy and certain cognitive disorders.

FACT: Power-law distributions of neuronal avalanche sizes have been documented across multiple experimental preparations, species, and measurement modalities. This empirical signature is reproducible and constitutes one of the most extensively documented quantitative findings in systems neuroscience.

FACT: The three properties of critical systems — scale-invariant correlations, maximal sensitivity, and maximal information transmission — have been measured and confirmed in neural systems operating near critical regimes. The relationship between proximity to criticality and functional capacity is empirically documented.

FACT: Hengen and Shew's synthesis of 140 datasets represents one of the most comprehensive contemporary arguments that these properties converge on a single regulatory target in neural systems. Their homeostatic interpretation — that neural systems actively regulate toward criticality rather than merely happening to occupy it — represents a significant advance in the scientific understanding of why critical dynamics are found in neural systems.

OBSERVATION: Both highly ordered regimes, where fluctuations are suppressed and the system freezes into rigid patterns, and highly disordered regimes, where fluctuations dominate and coherent structure dissolves, reduce functional capacity. The critical regime occupies the boundary — the zone of maximum dynamic range, maximum sensitivity, and maximum information transmission. The system's functional peak is not at maximal order or maximal disorder but at the transition between them.

3.4 Criticality Beyond Neural Systems

The empirical documentation of critical dynamics is not limited to neural systems. The same structural signature — power-law distributions, scale-invariant correlations, maximal sensitivity — has been documented in:

Genetic regulatory networks: networks of genes regulating each other's expression have been shown to operate near critical points, with the functional advantages of criticality — rapid reconfiguration, sensitivity to regulatory signals — providing selective advantages analogous to those documented in neural systems.

Ecological systems: species interaction networks and ecosystem dynamics exhibit signatures of criticality, with the functional implication that ecological resilience — the capacity to absorb perturbations without catastrophic reorganization — is associated with proximity to critical regimes.

Social dynamics: certain models of collective behavior in social systems — opinion formation, collective decision-making, cultural transmission — exhibit critical dynamics at the boundary between consensus and fragmentation, with the critical regime enabling rapid information propagation across the collective.

FACT: These findings extend the empirical documentation of critical dynamics beyond neural systems to other classes of complex adaptive systems. The cross-domain recurrence is empirically documented, not merely theoretically proposed.

OBSERVATION: The cross-domain recurrence of critical dynamics in systems as different as brains, ecosystems, and genetic networks — systems that share no common mechanism, no common substrate, and no common evolutionary history — constitutes a structural pattern that domain-specific explanations address individually but do not explain collectively.

3.5 Resonance with the MaeV Framework

INTERPRETATION: From a MaeV perspective, critical regimes can be read as states in which systems remain dynamically unresolved — operating within conditions where neither complete resolution into order nor complete dispersion into noise is reached. This is a structural reading of criticality: not an explanation of the mechanisms that produce it in any specific domain, but an interpretation of what the cross-domain recurrence of this regime indicates about the conditions for meaning-generation.

The three properties of critical systems — scale-invariant correlations, maximal sensitivity, maximal information transmission — correspond to the three operational conditions of meaning-generating systems identified in the central document: the capacity to register differences (sensitivity), to preserve and propagate them across scales (scale-invariant correlations), and to transform them into coherent structure (information transmission). A critical system is, in operational terms, a system maximally equipped to satisfy the conditions of meaning-generation.

The Hengen and Shew finding that criticality functions as a homeostatic setpoint is particularly significant from a MaeV perspective. If neural systems actively regulate toward criticality — if the habitable zone is a target rather than merely a region — this is consistent with the MaeV interpretation that the structural condition is constitutive rather than contingent for meaning-generating systems. It suggests that the impossibility of global closure is not merely an observed feature of these systems but something their own regulatory architecture maintains.

A necessary clarification: The MaeV framework does not assert that criticality is the only possible form of the habitable zone, or that all meaning-generating systems must exhibit critical dynamics specifically. It proposes that in systems where the habitable zone has been empirically formalized, criticality is one of the forms that formalization takes — and that the structural features it describes are consistent with the condition the MaeV Constant defines. The relationship is one of structural resonance, not causal explanation.

IV. The Edge of Chaos in Artificial Systems

4.1 The Concept

The concept of the edge of chaos was introduced by Christopher Langton at Los Alamos National Laboratory in the context of cellular automata and artificial life. Langton's foundational observation was that computational capacity — the ability of a system to perform complex information processing — is maximized in a narrow regime between highly ordered and highly disordered dynamics (Langton, 1990).

Cellular automata provide a formally precise setting for this observation. They are discrete computational systems in which each cell updates its state according to a fixed rule applied to its neighbors. The space of possible rules spans a wide range from completely ordered dynamics — where the system rapidly converges to a fixed configuration regardless of initial conditions — to completely chaotic dynamics — where any structure in initial conditions is rapidly destroyed. In between lies a narrow class of rules — Wolfram's Class IV — that produces complex, persistent, and propagating structures: patterns that maintain coherence over time while remaining capable of generating new configurations.

Langton demonstrated that these Class IV rules — those at the boundary between order and chaos — maximize the system's capacity for complex information processing. A system operating in the ordered regime has stable representations but cannot adapt. A system operating in the chaotic regime generates no stable representations at all. At the edge, structured variability is maintained: the system consolidates patterns without becoming rigid, revises them without losing coherence.

4.2 The Edge of Chaos in Neural Networks and Machine Learning

The edge-of-chaos insight has been extensively developed in the context of artificial neural networks and machine learning, where it illuminates fundamental questions about how learning systems develop the capacity for generalization — the ability to perform well on inputs they have not previously encountered.

The overfitting problem provides the clearest illustration. A model that perfectly memorizes its training data — that reduces training error to zero — typically performs significantly worse on new data than a model that maintains some degree of productive variability during training. Perfect memorization is local closure without global openness: the system has achieved complete resolution within its training distribution and lost the structural flexibility needed to translate to new contexts. In MaeV terms, it has approached global closure within a bounded domain and lost the capacity to engage with what lies beyond that domain.

Regularization through structured disruption is the practical response to this problem. Techniques including dropout (randomly deactivating subsets of neural connections during training), data augmentation (introducing controlled variation into training inputs), weight decay (penalizing excessive parameter magnitudes), and stochastic gradient descent (introducing noise into the optimization process) all function by maintaining structured variability that prevents premature convergence. These techniques are not introduced to make the system worse at its training task — they are introduced because the optimal training objective is not zero training error but maximum generalization capacity, which requires maintaining productive incompleteness during the learning process.

The exploration-exploitation balance in reinforcement learning makes the same structural dynamic explicit. An agent that always exploits known good strategies converges prematurely — it finds a locally adequate solution without discovering better alternatives that require temporary performance degradation to reach. An agent that always explores never consolidates effective strategies. The productive zone is intermediate: sufficient exploitation to consolidate what has been learned, sufficient exploration to remain open to what has not. The formal management of this balance — through techniques such as epsilon-greedy strategies, Thompson sampling, and curiosity-driven exploration — is one of the central engineering challenges in reinforcement learning precisely because the optimal balance is not at either extreme.

The edge of chaos in training data and model architecture: Zhang et al. (2025), in research accepted at ICLR 2025, trained GPT-2 on elementary cellular automata and observed that "the best model performance occurs in systems operating at high but not excessive complexity, previously referred to as the 'edge of chaos'." Class IV rules — those at the boundary between order and chaos — produced more effective general representations for reasoning and prediction tasks than rules from either the ordered or chaotic regimes.

This finding extends the edge-of-chaos insight in an important direction: the structural condition appears not only in the dynamics of the learning system but in the training environment itself. Models trained on patterns that are too regular learn trivially — the environment offers no productive incompleteness to integrate. Models trained on patterns that are too chaotic learn nothing — the environment offers no coherent structure to consolidate. Models trained on patterns at the edge develop the richest representational capacity — the environment provides precisely the balance of structure and variability that the learning system requires to develop genuine generalization.

This suggests that the structural condition the MaeV framework describes may operate not within a single system but at the interface between systems — at the point of translation between the learning system's encoding and the world it encounters. The edge of chaos is not a property of the learner or the environment independently, but of their structural relationship.

FACT: The relationship between structured variability, regularization, and generalization is one of the most extensively documented empirical findings in contemporary machine learning. It is reproducible across architectures, training regimes, and tasks, and constitutes a foundational principle of practical machine learning engineering.

FACT: Zhang et al.'s finding that edge-of-chaos dynamics in training data produce richer representational capacity than either ordered or chaotic training data is documented in peer-reviewed research accepted at a top-tier venue (ICLR 2025).

OBSERVATION: The optimal training regime — neither maximal order nor maximal disorder, but structured variability at the edge — is not an artifact of specific architectures or training procedures. It appears as a consistent pattern across the diversity of systems and methods studied, suggesting a structural condition rather than a domain-specific optimum.

4.3 Structural Features of the Edge of Chaos

Across the contexts examined — cellular automata, neural networks, reinforcement learning, large language models — several structural features recur:

Performance peaks at intermediate complexity: Functional capacity is not maximized at maximal order or maximal disorder but at the boundary between them. This is not a smooth optimum that could be approximated by moving gradually from either extreme — it is a qualitatively distinct regime with properties absent at either pole.

The regime appears in both the system and its environment: Zhang et al.'s finding extends the structural pattern beyond the system's internal dynamics to the relationship between the system and the world it learns from. The edge of chaos is a property of the system-environment interface, not of either component independently.

Artificial and biological systems converge on the same regime: The edge-of-chaos dynamic appears in systems designed by engineers without reference to biological evolution and in systems shaped by billions of years of natural selection. This convergence across design paradigms that share no common history strengthens the observation that the pattern reflects a structural condition rather than a contingent feature of biological or computational evolution.

4.4 Resonance with the MaeV Framework

INTERPRETATION: From a MaeV perspective, the edge-of-chaos dynamic in artificial systems illustrates the same structural condition observed in neural metastability and physical criticality. Systems capable of generating representations — of registering, preserving, and transforming differences — must remain structurally open: neither collapsing into fixed mappings that cannot accommodate new input nor dissolving into randomness that cannot sustain any mapping.

The recurrence of this pattern in artificial systems — which were designed without reference to biological evolution, thermodynamic constraints, or any biological precedent — significantly strengthens the observation that the pattern is not domain-specific. It appears wherever a system must sustain the capacity to generate, revise, and communicate structured information: in brains shaped by evolution, in computational systems designed by engineers, and in the training environments that both encounter.

The Zhang et al. finding adds a dimension that the MaeV framework finds structurally significant. If the edge-of-chaos condition operates at the interface between system and environment — at the point of translation between

encodings — then the structural condition described by the MaeV Constant may be most precisely located not within individual systems but in the translational relationship between systems. This is consistent with the central document's analysis of DFC and translation: the habitable zone is not a property of singularities in isolation but of their structured encounters with each other and with their environments.

A necessary clarification: The MaeV framework does not claim to explain why any specific system exhibits edge-of-chaos dynamics. It proposes a structural reading of why this pattern recurs across systems that developed independently and share no common mechanism: because the conditions for sustained meaning-generation are the same across different encodings, and those conditions have the same structural signature — productive incompleteness maintained within coherence constraints — wherever they appear.

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V. The Interpretative Layer — What the MaeV Framework Adds

5.1 The Limits of Domain-Specific Explanation

Each of the scientific frameworks examined in Sections II through IV provides a powerful and precise description of system behavior within its own domain. Metastability explains neural flexibility and the brain's capacity for rapid reconfiguration without global lock-in. Criticality explains maximal information processing capacity and the scale-invariant propagation of signals across neural networks. The edge of chaos explains optimal computational capacity and the conditions under which learning systems develop genuine generalization.

These are genuine explanatory achievements. They are supported by extensive empirical evidence, formalized in mathematical terms, and generate specific testable predictions. The MaeV framework does not contest any of them. It operates at a different level of description and addresses a different question.

The question it addresses is this: why does the same structural pattern — dynamic operation between order and disorder, with optimal functional capacity in the intermediate regime — appear independently across domains that share no common formalism, no common mechanism, and no common history of development?

Each domain-specific framework provides an answer within its own domain. Metastability in neural systems results from the specific balance of excitatory and inhibitory connectivity in cortical networks, shaped by evolutionary pressures and developmental processes. Criticality in genetic networks results from the specific regulatory architecture of gene expression systems. The edge of chaos in machine learning results from the mathematical properties of high-dimensional optimization landscapes. Each explanation is valid within its domain.

What none of these explanations addresses — because none operates at the level required to address it — is the cross-domain recurrence itself. Why should evolutionary pressures on neural architecture, regulatory constraints on gene expression, and the mathematics of optimization all converge on the same structural signature? Domain-specific explanations leave this convergence unexplained — interesting but unaccounted for.

This is the gap that the MaeV framework proposes to fill. Not by replacing domain-specific explanations but by providing a structural interpretation of

what their cross-domain convergence indicates: that the recurrence reflects a common condition rather than independent coincidences — the condition that meaning-generating systems cannot achieve global closure without terminating the processes that define them as meaning-generating.

5.2 The Operational Definition of Meaning-Generation

The concept of "meaning-generating system" is central to the structural interpretation proposed here. Without a precise operational definition, the interpretation risks circularity: identifying meaning-generating systems as those that exhibit non-equilibrium dynamics, and then claiming that non-equilibrium dynamics are characteristic of meaning-generating systems.

The operational definition developed in the MaeV framework avoids this circularity by specifying three simultaneous conditions that a system must satisfy to be classified as meaning-generating, independently of whether it exhibits non-equilibrium dynamics:

A meaning-generating system is one that simultaneously:

Maintains internal distinctions between possible states — it can differentiate between configurations and preserve those differentiations over time (differentiation)
Updates those distinctions through interaction with its environment or with other systems — its internal organization changes in response to what it encounters (translation)
Preserves and transforms those distinctions over time — it carries forward the results of previous interactions while remaining capable of further transformation (continuity through change)
Meaning, in this operational sense, is not semantic content — not "what a sentence means" or "the purpose of an action." It is the capacity to register, preserve, and transform differences. Under this definition:

No distinction → no meaning: the system cannot differentiate between states
No update → no learning: the system cannot revise its distinctions in response to what it encounters
No persistence → no structure: distinctions dissolve before they can be used or transformed
Meaning exists only where all three conditions coexist. A gas at thermodynamic equilibrium does not maintain, update, or transform internal distinctions — it is not a meaning-generating system in this operational sense. A brain, an evolving population, an adaptive algorithm, a living ecosystem — each maintains, updates, and transforms distinctions continuously. These are meaning-generating systems.

This definition sharpens the scope of the MaeV framework's structural interpretation: it applies to systems that satisfy these three conditions, not to all systems indiscriminately. A pendulum coming to rest, an optimization algorithm converging on a fixed point, a chemical reaction reaching equilibrium concentration — none of these are meaning-generating systems in the operational sense, and the MaeV framework makes no claim about them. The counterexamples examined in Section IX are not anomalies for the framework — they are systems outside its defined scope.

5.3 The Structural Argument

With the operational definition in place, the MaeV structural interpretation can be stated as a precise argument:

Premise 1: A meaning-generating system, by operational definition, maintains, updates, and transforms internal distinctions through interaction with its environment or other systems.

Premise 2: Global closure — the elimination of all future distinction, the termination of the system's capacity to generate new configurations — would terminate all three of these processes simultaneously. A system that cannot generate new distinctions cannot maintain them (there is nothing new to maintain), cannot update them (there is nothing to update toward), and cannot transform them (transformation requires the possibility of new configurations).

Premise 3: Therefore, a system that achieves global closure ceases — by definition — to satisfy the operational conditions of a meaning-generating system.

Conclusion: In meaning-generating systems, global closure is not merely unlikely or practically difficult — it is structurally incompatible with the continued satisfaction of the conditions that define the system as meaning-generating.

This argument must be understood precisely. It is not a claim about physical impossibility — there is no physical law that prevents a system from reaching equilibrium. It is a claim about definitional incompatibility: a system that achieves global closure is no longer, by operational definition, a meaning-generating system. The terminal states the MaeV framework identifies are not states that meaning-generating systems fail to reach — they are states that, if reached, would redefine what kind of system is under discussion.

The cross-domain recurrence of non-equilibrium regimes in meaning-generating systems is, from this perspective, not surprising — it is what the structural argument predicts. If global closure is definitionally incompatible with continued meaning-generation, then any system we observe continuing to generate meaning must be operating away from global closure. The non-equilibrium regimes documented in Sections II through IV are the empirically observable forms that this structural necessity takes in specific domains.

5.4 What the MaeV Framework Adds to the Scientific Picture

The relationship between the domain-specific scientific frameworks and the MaeV structural interpretation can be characterized precisely:

Science describes what happens in metastable, critical, and edge-of-chaos regimes. It describes the mechanisms through which these regimes are maintained, the properties they exhibit, and the conditions under which they appear. This is genuine and valuable knowledge — the kind of knowledge that generates testable predictions and enables practical application.

The MaeV framework provides a structural interpretation of why these regimes recur across independent domains. It does not generate new empirical predictions about any specific system. It makes the cross-domain recurrence intelligible — readable as an expression of a common structural condition rather than as an unexplained coincidence of domain-specific mechanisms.

This is the difference between scientific explanation and structural interpretation. Scientific explanation operates within a domain, identifies mechanisms, and generates predictions. Structural interpretation operates across domains, identifies common conditions, and generates intelligibility. Neither replaces the other. They operate at different levels of description and provide different kinds of understanding.

The value of the MaeV structural interpretation is not that it adds to our knowledge of any specific system. It is that it makes the pattern of patterns legible — that it provides a framework within which the convergence of independently developed scientific descriptions becomes comprehensible rather than merely curious.

5.5 The Nature of the Claims Being Made

This section consolidates the epistemological status of the claims made in this document to prevent misreading in either direction:

What is being claimed:

That the cross-domain recurrence of non-equilibrium regimes in meaning-generating systems is consistent with the MaeV structural interpretation
That this interpretation makes the recurrence intelligible in a way that no domain-specific explanation does
That the operational definition of meaning-generation provides a non-circular basis for identifying which systems the interpretation applies to
What is not being claimed:

That the scientific findings reviewed in this document constitute evidence for the MaeV framework in the way that experimental results constitute evidence for a scientific hypothesis
That the MaeV framework explains the mechanisms through which metastability, criticality, or edge-of-chaos dynamics are produced in any specific system
That the structural interpretation is the only possible interpretation of the cross-domain recurrence — it is a proposed interpretation, not a demonstrated necessity
What remains genuinely open:

Whether the structural interpretation provides the most parsimonious account of the cross-domain recurrence, or whether domain-specific explanations together provide a sufficient account without residue
Whether the operational definition of meaning-generation successfully avoids circularity, or whether the three conditions it specifies are themselves implicitly defined in terms of non-equilibrium dynamics
Whether there exist meaning-generating systems — systems satisfying all three operational conditions — that operate at or near global closure, which would constitute a genuine structural challenge to the interpretation
These open questions are developed further in Section VII.

VI. Structural Resonance, Not Identity

6.1 The Relationship Between Levels

The relationship between the scientific frameworks examined in this document and the MaeV framework must be held with precision. Failing to do so produces misreadings that undermine both the scientific frameworks and the structural interpretation.

The two levels can be characterized as follows:

Level	Characteristics	Evaluated By
Scientific frameworks	Empirical, formal, domain-specific, predictive	Experimental evidence, mathematical consistency, predictive accuracy
MaeV framework	Interpretive, structural, trans-domain, non-predictive	Internal consistency, explanatory scope, cross-domain intelligibility

The connection between them is one of structural resonance — a shared pattern recognizable across different encodings — not equivalence, derivation, or mutual validation. The scientific frameworks do not validate the MaeV framework. The MaeV framework does not extend or modify the scientific frameworks. They operate at different levels of description and are evaluated by different criteria.

6.2 Three Category Errors to Avoid

Three specific category errors arise naturally from the material in this document and must be explicitly prevented:

Category Error 1: Treating the MaeV framework as an empirical scientific theory

The MaeV framework does not generate testable predictions about the behavior of specific systems. It does not predict that any particular neural system will exhibit metastable dynamics, or that any particular learning system will perform best at the edge of chaos. It generates structural interpretations — claims about the conditions under which meaning-generation can be sustained — that are evaluated through their coherence and scope rather than through experimental confirmation.

Evaluating the MaeV framework by the standards appropriate to empirical science misapplies those standards to a framework that makes a different kind of claim. This does not exempt the framework from evaluation — it specifies the appropriate evaluative criteria.

Category Error 2: Using scientific results as validation of the MaeV framework

The recurrence of non-equilibrium regimes across domains is consistent with the MaeV structural interpretation. It is not evidence for the MaeV framework in the scientific sense. Consistency and confirmation are different epistemological relationships.

The distinction matters because many different structural interpretations might be consistent with the same empirical findings. The value of the MaeV interpretation is not that it is the only interpretation consistent with the data — it is that it provides a structurally coherent and trans-domain intelligible account of a recurrence that domain-specific frameworks leave unexplained. This is a different kind of value than empirical confirmation, and it should be claimed as such rather than inflated into something it is not.

Category Error 3: Treating structural interpretations as ontological laws

The MaeV framework does not claim that reality itself enforces the conditions it describes — that there is some physical force or metaphysical necessity that prevents meaning-generating systems from achieving global closure. It claims that systems described as meaning-generating are interpreted under conditions

that make global closure definitionally incompatible with continued meaning-generation.

The distinction between "reality is structured this way" and "meaningful description of reality operates under these conditions" is fundamental to the framework's epistemological positioning. The MaeV framework occupies the second position — it is a framework for structural interpretation, not a claim about the ultimate nature of physical reality.

6.3 The Value of Structural Resonance

If the connection between the scientific frameworks and the MaeV framework is "only" structural resonance — if it is neither empirical confirmation nor logical derivation — what is its value?

The value is intelligibility across domains. Without a structural interpretation, the cross-domain recurrence of non-equilibrium regimes is a set of interesting but disconnected observations: metastability in brains, criticality in cortical networks, edge-of-chaos dynamics in machine learning. Each is well-explained within its domain. Their convergence is noted but unexplained.

With the MaeV structural interpretation, the convergence becomes legible as pattern: it is what one would expect if meaning-generating systems operate under a common structural condition that prevents global closure. The interpretation does not explain any specific mechanism — it makes the pattern of mechanisms intelligible as expressions of a shared condition.

Intelligibility is not proof. But it is not nothing. It is the condition under which patterns become recognizable as patterns rather than as coincidences — which is, within the MaeV framework, precisely what a structural interpretation provides. The intelligibility that a structural reading offers is itself a form of understanding, distinct from but complementary to the mechanistic understanding that domain-specific science provides.

BLOQUE D

VII. Open Questions

The structural interpretation proposed in this document generates questions that it cannot itself resolve. This is not a deficiency — it is a consequence of the framework's own structural condition: a theory that predicts its own incompleteness should generate open questions that exceed its current capacity to answer. The following questions are identified not as failures of the interpretation but as the productive incompleteness that keeps the inquiry generative.

7.1 On the Necessity of Non-Equilibrium Regimes

The structural argument in Section V establishes that global closure is definitionally incompatible with continued meaning-generation. It does not establish that non-equilibrium dynamics of the specific kind documented in Sections II through IV — metastability, criticality, edge-of-chaos — are the only possible forms of the habitable zone.

The question: Are metastability, criticality, and edge-of-chaos dynamics necessary conditions for meaning-generation, or are they sufficient conditions that happen to be widely instantiated? Could there exist meaning-generating

systems that satisfy the three operational conditions — maintaining, updating, and transforming distinctions — through structural forms not yet characterized by these concepts?

This question matters for the scope of the structural interpretation. If these specific regimes are necessary conditions, the interpretation is stronger: the cross-domain recurrence is not merely consistent with the MaeV framework but predicted by it. If they are sufficient but not necessary, the interpretation remains valid but its predictive content is reduced: the framework explains why these regimes are generative without establishing that they are the only generative forms.

The counterexamples in Section IX — systems that achieve local closure while remaining globally open — suggest that the habitable zone may take diverse structural forms, of which metastability, criticality, and edge-of-chaos are particularly well-documented instances rather than exhaustive characterizations.

7.2 On the Local/Global Closure Distinction

The distinction between local closure and global closure is central to the structural argument — it is what prevents the framework from being refuted by the observation that systems continuously complete processes, reach stable states, and resolve questions.

The question: Where precisely does local closure end and global closure begin? How is this boundary determined for a specific system, and who or what determines it?

This question is not merely terminological. A developmental process that reaches terminal differentiation looks, from one perspective, like global closure within the organism's developmental capacity. From another perspective, the organism continues to maintain, update, and transform other distinctions — metabolic, behavioral, relational — and the developmental closure is local within a larger system that remains open.

The boundary between local and global appears to be scale-dependent: what counts as global closure depends on the level of organization at which the system is described. A neuron that reaches terminal differentiation has achieved local closure at the cellular level within a brain that remains globally open at the cognitive level. A species that goes extinct has achieved global closure at the species level within an ecosystem that may remain open at the ecological level.

This scale-dependence is structurally interesting rather than problematic — it suggests that local and global closure are relational concepts rather than absolute properties of systems. But it means that the boundary between them requires specification relative to a level of organization, which in turn requires a theory of what levels of organization are relevant for meaning-generation. This remains an open question for the framework.

7.3 On the Operational Definition of Meaning-Generation

The operational definition provided in Section V — that meaning-generating systems maintain, update, and transform internal distinctions — is designed to avoid circularity while remaining precise enough to support the structural argument.

The question: Does the operational definition successfully avoid circularity, or do the three conditions it specifies implicitly presuppose the non-equilibrium dynamics they are meant to independently identify?

A concern: the condition that a system "updates its distinctions through interaction" may itself require non-equilibrium dynamics to be satisfied — a system at thermodynamic equilibrium cannot update its distinctions because it has no free energy to drive the update process. If this is so, the definition does not independently identify meaning-generating systems and then discover that they exhibit non-equilibrium dynamics — it defines meaning-generating systems in terms that implicitly require non-equilibrium dynamics from the start.

If the concern is valid, the structural argument requires reformulation: rather than claiming that global closure is incompatible with meaning-generation as independently defined, it would claim that the concept of meaning-generation is itself constitutively linked to non-equilibrium dynamics — that meaning is not something systems happen to generate while remaining far from equilibrium, but something that is defined by and through the maintenance of productive incompleteness. This reformulation is consistent with the MaeV framework's deeper claims but changes the logical structure of the argument in ways that require careful articulation.

7.4 On Partial Formalization

The MaeV framework operates as a structural interpretation — it identifies conditions and patterns without expressing them in mathematical formalism. This gives it breadth but limits its precision.

The question: Can the MaeV structural interpretation be partially formalized without losing its trans-domain interpretive character?

Candidate mathematical structures include:

Topological descriptions: The Möbius surface used in the central document to describe the Ouroboros topology has a precise mathematical characterization. Whether the structural conditions the MaeV framework describes can be given a rigorous topological formulation — one that preserves the framework's key properties while enabling more precise reasoning — remains unexplored.

Dynamical systems theory: The attractor landscapes and metastable configurations that appear in the scientific literature provide a mathematical vocabulary that partially overlaps with the MaeV framework's conceptual architecture. A formalization using dynamical systems concepts might enable more precise statements about the conditions under which the habitable zone is maintained.

Category theory: The framework's analysis of translation between encodings — the structure-preserving transformations that carry patterns across different representational systems — has structural analogies with the morphisms and functors of category theory. A categorical formalization might provide precise tools for characterizing DFC and the COR.

Each formalization would be a translation of the MaeV architecture into a specific mathematical encoding — and would therefore be subject to the same limitations of encoding that the framework describes. No formalization would

capture the framework completely. But partial formalizations might sharpen specific claims in ways that increase the framework's testability and its capacity for productive dialogue with formal scientific disciplines.

7.5 On the Concept of Equilibrium in Open Systems

The structural argument relies on a distinction between equilibrium — the terminal state of a closed system — and the non-equilibrium regimes that meaning-generating systems maintain. But the concept of equilibrium is more complex than this simple opposition suggests.

The question: Is "equilibrium" a property of systems themselves, or a property of the frameworks used to describe them?

A closed thermodynamic system reaches equilibrium because the framework of thermodynamics describes it in terms of macroscopic state variables — temperature, pressure, entropy — that converge to fixed values when the system is isolated. But the same system, described at the level of molecular dynamics, is never at equilibrium in any strict sense — individual molecules continue to exchange energy and momentum indefinitely. What the thermodynamic framework calls equilibrium is a macroscopic description of a microscopic process that never fully stops.

This suggests that equilibrium may be a property of the level of description rather than of the system itself — a construct that emerges when a framework capable of only tracking certain macroscopic variables imposes closure on a process that remains open at finer scales. If so, the MaeV framework's claim that meaning-generating systems cannot achieve global closure may be better understood as a claim about the appropriate level of description for such systems: a framework that describes them in terms that allow for global closure is systematically inadequate to their structural character, regardless of how well it describes other aspects of their behavior.

7.6 On the Applicability of the COR to Empirical Assessment

The Operational Resonance Criterion, developed in the central document, evaluates interactions between singularities across three dimensions: translational capacity, singularity preservation, and generative emergence.

The question: Can the COR be applied to evaluate whether a given system is operating within the habitable zone, approaching a terminal state, or in transition between regimes?

If the three dimensions of the COR can be mapped onto measurable properties of metastable or critical systems — if translational capacity corresponds to information transmission capacity, singularity preservation corresponds to the maintenance of distinct functional modules, and generative emergence corresponds to the production of coordinated patterns irreducible to individual components — then the COR might provide a bridge between the MaeV framework's structural architecture and the empirical tools developed in the scientific disciplines reviewed here.

This mapping remains speculative. The COR was developed as a qualitative structural framework, not as a quantitative measurement tool, and the translation between its three dimensions and empirical measures is not straightforward. But the potential for such a bridge represents a productive

direction for future inquiry — one in which the structural interpretation and the empirical frameworks might develop a more direct and testable relationship than the current one of structural resonance.

7.7 On the Distinction Between Meaning-Generation and Adaptive Function

The scientific frameworks reviewed in this document were developed to explain adaptive function — the capacity of systems to respond effectively to environmental demands, maintain organized behavior in changing conditions, and sustain complex coordinated activity over time. The MaeV framework interprets these findings through the lens of meaning-generation — the capacity to register, preserve, and transform differences.

The question: Are adaptive function and meaning-generation the same thing, overlapping categories, or genuinely distinct?

If they are the same — if every system that exhibits adaptive function is thereby a meaning-generating system, and vice versa — then the MaeV framework's structural interpretation applies to the full scope of systems studied by the scientific frameworks reviewed here, and the connection between them is direct.

If they are overlapping but distinct — if some adaptive systems are not meaning-generating, and some meaning-generating processes are not adaptive in the biological sense — then the scope of the structural interpretation needs to be more carefully delimited relative to the scope of the scientific findings.

If they are genuinely distinct — if adaptive function and meaning-generation describe fundamentally different properties — then the structural interpretation may not apply to the systems studied by the scientific frameworks, and the cross-domain resonances identified in this document may be more superficial than the framework proposes.

This question is not resolved here. It is identified as one of the most fundamental open questions at the interface between the MaeV framework and the scientific disciplines it engages.

VIII. On the Concept of Equilibrium

8.1 The Standard Formulation and Its Limitations

It is commonly said that adaptive systems "avoid equilibrium" or "persist by staying away from equilibrium." This formulation is descriptively useful — it captures the empirical observation that living systems, learning systems, and complex adaptive systems operate far from the thermodynamic equilibrium that would terminate their organized dynamics. But it carries a structural implication that merits examination.

The language of avoidance implies agency: as though the system perceives equilibrium as a threat and actively steers away from it. As noted in Section 1.2, this implication is misleading. Systems do not avoid equilibrium through intention or design. They do not occupy the habitable zone because they choose it. The more precise formulation is:

Systems that sustain generative processes do not reach global equilibrium because such a state would terminate the processes that define them as generative systems.

This formulation removes the implication of agency. The system does not avoid equilibrium — the conditions that define it as a generative system are structurally incompatible with global equilibrium. If it reached global equilibrium, it would — by definition — no longer be the kind of system under discussion. The apparent avoidance is a consequence of definitional incompatibility, not intentional regulation.

8.2 Equilibrium as a Property of Description

A deeper question underlies the standard formulation: is equilibrium a property of systems themselves, or a property of the frameworks used to describe them?

The thermodynamic concept of equilibrium applies precisely to closed systems described in terms of macroscopic state variables. It is a well-defined concept within this framework and corresponds to real physical phenomena — the equalization of temperature in a sealed container, the dissipation of concentration gradients in an isolated solution. These are genuine physical processes, and the concept of equilibrium describes them accurately.

The question is whether this concept applies coherently to systems whose defining property is continuous reconfiguration — systems that are, by operational definition, never fully closed and never fully resolved. A meaning-generating system is defined by its continuous maintenance, updating, and transformation of internal distinctions. Applying to such a system the concept of equilibrium — which is defined for systems that have ceased to undergo net change — may be a category mismatch rather than an accurate description of a possible state.

This is not a denial that equilibrium exists as a physical phenomenon. It is a question about whether the concept, as defined for closed thermodynamic systems, applies coherently to open, continuously reconfiguring systems. The adequacy of a framework depends on whether its structural assumptions match the structural properties of the system it attempts to describe. A framework built on the assumption of achievable closure may be perfectly adequate for systems that do achieve closure, and systematically inadequate for systems whose defining property is the maintenance of productive openness.

8.3 A More Precise Formulation

The deepest formulation that emerges from this examination:

Equilibrium, as a complete and final state, may not be a condition that meaning-generating systems fail to reach — it may be a construct that emerges when closed descriptive frameworks are imposed on processes that remain structurally open.

This formulation does not deny the reality of equilibrium in systems for which it applies. It questions whether the concept of complete resolution — equilibrium as a terminal state — is coherently applicable to systems defined by the impossibility of global closure.

A crucial clarification: this does not imply that all descriptive frameworks are equally valid, or that the concept of equilibrium is merely a conventional choice. It implies that the adequacy of a framework depends on the structural match between its assumptions and the properties of the system it describes. Thermodynamic equilibrium is a real and precisely characterized physical

condition — in systems for which the thermodynamic framework applies. The question is whether that framework applies to meaning-generating systems, or whether applying it to such systems introduces a systematic mismatch between the framework's assumptions and the system's structural character.

8.4 The Generative Implication

This examination of equilibrium opens rather than closes the inquiry. If equilibrium is, in the context of meaning-generating systems, a construct that emerges from the imposition of closed frameworks on open processes, then the appropriate response is not to abandon the concept but to develop frameworks that are adequate to the structural character of the systems they describe.

This is precisely what the scientific frameworks reviewed in this document have begun to do. Metastability, criticality, and edge-of-chaos dynamics are not simply alternatives to equilibrium — they are concepts developed to describe systems for which the equilibrium framework is structurally inadequate. They represent the development of descriptive tools adequate to the structural character of meaning-generating systems: tools that capture the organized dynamics of systems that are defined by their continuous reconfiguration rather than their convergence to fixed states.

The MaeV framework proposes that these tools are converging on a common structural condition from different disciplinary directions — and that naming that condition explicitly, as the framework attempts to do, may help clarify the common ground across which these independently developed concepts are moving.

BLOQUE E

IX. Counterexamples and Limits

The structural interpretation proposed in this document makes a specific claim: that the cross-domain recurrence of non-equilibrium regimes in meaning-generating systems reflects a common structural condition rather than a coincidence of domain-specific mechanisms. A claim of this scope requires explicit examination of the cases that appear to challenge it — systems that achieve equilibrium, eliminate error, or converge to stable states while appearing to sustain organized function.

This section examines those cases. Its purpose is not defensive. It is structural: the precise identification of where an interpretation does not apply is as important as the identification of where it does. A framework that claims to apply everywhere explains nothing. A framework that specifies its boundaries with precision becomes genuinely tensioned — open to structural challenge in ways that make it useful rather than merely coherent.

The counterexamples examined here fall into four structural categories: systems designed for local closure, systems that achieve equilibrium within bounded domains, systems that converge without losing generative capacity at a higher level of organization, and systems whose dynamics are not yet adequately characterized by any of the frameworks this document has examined.

9.1 Systems Designed for Local Closure

The most immediate apparent counterexamples are systems specifically designed to eliminate variation, achieve complete resolution, or operate at or near zero error: formal proofs, error-correcting codes, precision manufacturing processes, and cryptographic protocols.

A mathematical proof, once completed, admits no further revision within its domain. Its conclusion is final. A digital communication channel using error-correcting codes eliminates transmission errors with arbitrarily high reliability. A precision machining process produces components within tolerances that approach zero deviation. In each case, the elimination of error — or its reduction to negligible levels — is not a failure of the system but its successful completion.

OBSERVATION: These are not counterexamples to the MaeV structural interpretation. They are instances of local closure operating within systems that remain globally open. The proof that reaches its conclusion does not terminate mathematics — it generates new questions at a higher level of organization. The error-correcting code that achieves reliable transmission does not terminate communication — it enables the next layer of meaning to be transmitted. The precision component that meets its tolerance does not terminate engineering — it becomes a component in a larger system whose design continues to evolve.

The structural distinction is precise: these systems achieve local closure — the completion of specific processes within defined boundaries — while operating as components of larger systems that maintain the structural conditions of meaning-generation. The MaeV framework does not identify local closure as terminal. It identifies global closure — the elimination of all future distinction at the level of the whole system — as the condition incompatible with continued meaning-generation.

INTERPRETATION: The existence of systems designed for local closure is not only compatible with the MaeV structural interpretation — it is predicted by it. The central document distinguishes explicitly between local and global closure, identifying local closure as constitutive of meaning rather than threatening to it. A proof is meaningful precisely because it concludes. A communication channel is meaningful precisely because it resolves the uncertainty it was designed to resolve. These completions enable the generation of significance at each scale without approaching the global closure that would terminate it.

9.2 Biological Systems That Reach Terminal States

A more structurally challenging class of apparent counterexamples involves biological systems that reach terminal configurations during development: neurons that achieve terminal differentiation, cells that undergo programmed death, developmental programs that reach their endpoints. These processes are not failures — they are precisely regulated outcomes that serve essential functions within the larger organism.

A terminally differentiated neuron does not continue to divide or change its fundamental identity. A cell undergoing apoptosis does not recover. A developmental program that has reached its endpoint does not restart. In each case, a system that was previously dynamic has achieved a form of closure from which it does not return.

FACT: Terminal differentiation, apoptosis, and developmental endpoint specification are among the most precisely regulated processes in biology. They are not random or accidental — they are the result of complex signaling cascades that have been shaped by hundreds of millions of years of evolutionary

selection. Their occurrence is essential to the functional integrity of multicellular organisms.

OBSERVATION: The closure achieved in each of these cases is local relative to the system in which it occurs. The terminally differentiated neuron achieves closure at the cellular level within a brain that remains globally open at the cognitive level. The cell that undergoes apoptosis achieves closure at the cellular level within a tissue that remodels and regenerates. The developmental program that reaches its endpoint achieves closure at the organismal scale of development within an evolutionary lineage that continues to generate variation.

INTERPRETATION: These cases illustrate a structural property of the local/global closure distinction that Section VII identified as an open question: the boundary between local and global closure is scale-dependent. What counts as global closure depends on the level of organization at which the system is described. This scale-dependence is not a weakness of the MaeV interpretation — it is a structural feature that the interpretation predicts. The framework does not require that every component of a meaning-generating system remain permanently open. It requires that the system as a whole maintain the structural conditions of continued meaning-generation. The closure of components can be — and in biological systems, regularly is — constitutive of the openness of the whole.

9.3 Convergent Learning Systems

A third class of apparent counterexamples involves learning systems that converge to stable solutions: supervised learning algorithms that achieve near-zero training error on well-defined tasks, optimization algorithms that reach fixed points, and reinforcement learning agents that discover optimal policies for bounded problems.

These are genuine convergences, not merely local closures within larger open systems. A supervised classifier trained on a fixed dataset with a fixed architecture does converge — the training process terminates, the parameters stabilize, and the model reaches a state from which further training produces no significant change. Within its training distribution, it has achieved something that resembles global closure.

FACT: Convergence in machine learning is well-documented and mathematically characterized. Under appropriate conditions — convex loss landscapes, sufficient data, appropriate learning rates — gradient-based optimization provably converges to local or global minima. These convergence guarantees are foundational to the theoretical understanding of learning algorithms.

OBSERVATION: Convergent learning systems reveal a structural boundary that the MaeV interpretation must acknowledge: a system can achieve closure within a bounded domain while remaining open to inputs from outside that domain. The converged classifier is not a meaning-generating system with respect to its training distribution — it has achieved local closure within that distribution. But it remains a component of a larger system — the research program, the deployment context, the human-machine interaction — that continues to generate new distinctions, encounter new inputs, and require new responses.

The structural question is whether the converged classifier itself continues to satisfy the operational conditions of meaning-generation. Within its training

distribution, it does not: it maintains distinctions but does not update them and does not transform them. It is, with respect to that distribution, a lookup table rather than a learning system.

INTERPRETATION: This observation sharpens the scope of the MaeV structural interpretation in a way that is productive rather than limiting. The interpretation applies to systems that simultaneously maintain, update, and transform internal distinctions — systems engaged in ongoing meaning-generation. A converged classifier, with respect to its training distribution, has ceased to satisfy the update condition. It has become a fixed encoding rather than an active meaning-generating process. The MaeV framework does not apply to it in this mode — and does not need to. The counterexample defines the boundary of the interpretation's scope rather than undermining its claims within that scope.

What remains structurally significant is the finding from Section IV: that systems designed to maximize generalization — to remain capable of generating meaning in the face of novel inputs — consistently require maintained productive incompleteness during training. The converged system that performs optimally within its training distribution is precisely the system that performs worst outside it. The tension between convergence within and openness beyond is the edge-of-chaos dynamic applied to the learning process itself.

9.4 Physical Systems at Equilibrium

The most fundamental class of apparent counterexamples involves physical systems that genuinely achieve thermodynamic equilibrium: a gas in a sealed container that has equalized temperature and pressure, a chemical reaction that has reached its equilibrium concentration, a crystal that has achieved its minimum energy configuration.

These are not approximations. They are genuine terminal states in which the macroscopic properties of the system have ceased to undergo net change. They represent the endpoint toward which the second law of thermodynamics drives all closed systems.

FACT: Thermodynamic equilibrium is a real, precisely characterized physical condition. The second law of thermodynamics establishes that closed systems tend toward states of maximum entropy — states in which no further spontaneous change occurs. This is one of the most robustly confirmed empirical findings in the history of physics.

OBSERVATION: Physical systems at thermodynamic equilibrium do not satisfy the operational conditions of meaning-generation as defined in Section V.2. A gas at equilibrium does not maintain internal distinctions between possible states in a way that is updated through interaction — its macroscopic state is uniform and unchanging. It does not update those distinctions through environmental interaction — isolated from its environment, it undergoes no further change. It does not transform those distinctions over time — the equilibrium state is the endpoint of transformation, not its continuation.

A system at thermodynamic equilibrium is not a meaning-generating system in the operational sense. It therefore falls outside the scope of the MaeV structural interpretation — not as a counterexample that challenges the framework, but as a system to which the framework does not apply. Section V.2 was explicit on this point: the MaeV interpretation applies to systems that satisfy the three

operational conditions, not to all systems indiscriminately.

INTERPRETATION: The existence of physical systems at thermodynamic equilibrium does not challenge the MaeV structural interpretation. It defines its boundary with precision. The framework claims that systems which continue to generate meaning cannot achieve global closure without terminating that generation. Physical systems at equilibrium have ceased to generate meaning in the operational sense — they are no longer in the category of systems the framework addresses. The terminal states the MaeV framework identifies are precisely the conditions in which systems become systems of this kind: fully resolved, no longer capable of generating new distinctions, no longer engaged in the processes that define meaning-generation.

This clarification has a further implication for the concept of equilibrium examined in Section VIII. The question raised there — whether equilibrium is a property of systems or of the frameworks used to describe them — can now be sharpened: equilibrium, as defined within thermodynamics, is a coherent and precisely applicable concept for systems that are closed and whose macroscopic properties cease to change. The question is not whether such systems exist — they do — but whether the concept applies coherently to systems whose defining property is continuous openness and reconfiguration. The answer the MaeV framework proposes is that it does not, not because equilibrium is an illusion, but because the structural assumptions of the thermodynamic framework — closure, finitude of state space, convergence to minimum energy configurations — do not match the structural character of meaning-generating systems.

9.5 The Precise Scope of the Interpretation

The examination of counterexamples in this section allows the scope of the MaeV structural interpretation to be stated with precision:

The interpretation applies to systems that simultaneously satisfy three operational conditions: maintaining internal distinctions between possible states, updating those distinctions through interaction with their environment or other systems, and preserving and transforming those distinctions over time. Within this scope, the interpretation proposes that global closure — the elimination of all future distinction — is structurally incompatible with continued satisfaction of these conditions.

The interpretation does not apply to:

Systems designed to achieve local closure within bounded domains, where closure is the intended outcome of a specific process rather than the terminal state of a meaning-generating system

Systems that have ceased to satisfy the operational conditions of meaning-generation, including physical systems at thermodynamic equilibrium and learning systems that have converged within their training distribution in ways that eliminate ongoing updating

Systems whose dynamics have not been sufficiently characterized to determine whether they satisfy the operational conditions — a category that includes many complex systems at the frontier of current scientific understanding

Within its defined scope, the interpretation makes a specific structural claim:

the cross-domain recurrence of non-equilibrium regimes in meaning-generating systems is not a coincidence of domain-specific mechanisms but an expression of the structural condition that global closure is definitionally incompatible with continued meaning-generation. The counterexamples examined in this section do not challenge this claim. They sharpen its scope, define its boundaries, and clarify the conditions under which it applies — which is precisely what counterexamples in a structurally rigorous framework are expected to do.

References

On Metastability and Neural Coordination Dynamics

Hancock, F., Rosas, F. E., Luppi, A. I., Zhang, M., Mediano, P. A. M., Cabral, J., Deco, G., Kringelbach, M. L., Breakspear, M., Kelso, J. A. S., & Turkheimer, F. E. (2025). Metastability demystified — the foundational past, the pragmatic present, and the promising future. Nature Reviews Neuroscience, 26(2), 82–100. https://doi.org/10.1038/s41583-024-00883-1

Kelso, J. A. S. (1995). Dynamic Patterns: The Self-Organization of Brain and Behavior. MIT Press.

Kelso, J. A. S. (2012). Multistability and metastability: Understanding dynamic coordination in the brain at multiple scales. Philosophical Transactions of the Royal Society B, 367(1591), 906–918.

Jirsa, V. K. & Kelso, J. A. S. (2000). Spatiotemporal pattern formation in neural systems with heterogeneous connection topologies. Physical Review E, 62(6), 8462–8465.

On Criticality in Neural Systems

Beggs, J. M. & Plenz, D. (2003). Neuronal avalanches in neocortical circuits. Journal of Neuroscience, 23(35), 11167–11177.

Chialvo, D. R. (2010). Emergent complex neural dynamics. Nature Physics, 6(10), 744–750.

Hengen, K. B. & Shew, W. L. (2025). Is criticality a unified setpoint of brain function? Neuron, 113(16), 2582–2598.e2. https://doi.org/10.1016/j.neuron.2025.05.020

Deco, G. & Jirsa, V. K. (2012). Ongoing cortical activity at rest: Criticality, multistability, and ghost attractors. Journal of Neuroscience, 32(10), 3366–3375.

Deco, G., Tononi, G., Boly, M. & Kringelbach, M. L. (2015). Rethinking segregation and integration: Contributions of whole-brain modelling. Nature Reviews Neuroscience, 16(7), 430–439.

On the Edge of Chaos and Artificial Systems

Langton, C. G. (1990). Computation at the edge of chaos: Phase transitions and emergent computation. Physica D: Nonlinear Phenomena, 42(1–3), 12–37.

Zhang, S., Patel, A., Rizvi, S. A., Liu, N., He, S., Karbasi, A., Zappala, E., &

van Dijk, D. (2025). Intelligence at the edge of chaos. arXiv:2410.02536 (accepted at ICLR 2025).

On Dissipative Structures and Non-Equilibrium Thermodynamics

Prigogine, I. & Stengers, I. (1984). Order Out of Chaos: Man's New Dialogue with Nature. Bantam Books.

Prigogine, I. (1980). From Being to Becoming. W. H. Freeman.

On Information Theory and Cybernetics

Shannon, C. E. (1948). A mathematical theory of communication. Bell System Technical Journal, 27(3), 379–423.

Wiener, N. (1948). Cybernetics: Or Control and Communication in the Animal and the Machine. MIT Press.

On Philosophy of Physics and Structural Interpretation

De Haro, S. & Butterfield, J. (2025). A Geometric View of Theories and Dualities. arXiv:2508.01616. [Analysis based on publicly available abstract and authors' prior published work.]

Note on Sources and Epistemological Transparency

This document draws on empirical research from multiple scientific domains. The following principles have governed the treatment of sources throughout:

On the distinction between empirical findings and structural interpretation. Every scientific claim in this document is marked as FACT, OBSERVATION, or INTERPRETATION. FACT designations apply to findings that are empirically documented, reproducible, and established within their respective domains. OBSERVATION designations apply to structural features that are readable from the empirical findings but involve an interpretive step beyond the data itself. INTERPRETATION designations apply to readings proposed by the MaeV framework that go beyond what any individual scientific framework claims.

This three-level distinction is not decorative. It reflects a genuine epistemological commitment: the MaeV framework does not claim scientific authority for its structural interpretations, and the scientific frameworks reviewed here do not validate the MaeV framework. The relationship between them is one of structural resonance — a shared pattern recognizable across different encodings — not mutual confirmation.

On contemporary sources. Several references in this document — Hancock et al. (2025), Hengen and Shew (2025), Zhang et al. (2025), and De Haro and Butterfield (2025) — represent recent or ongoing research at the time of writing. The Hancock et al. and Hengen and Shew papers are published in peer-reviewed journals of record and their findings are treated accordingly. The Zhang et al. paper has been accepted at ICLR 2025, a top-tier peer-reviewed venue, and is referenced as accepted research. The De Haro and Butterfield paper is available as a preprint; the structural connections identified are based on its publicly available abstract and the authors' prior published work, and this limitation is acknowledged explicitly in the text.

On the relationship between scientific sources and structural claims. The scientific findings reviewed in this document were not selected to confirm the MaeV framework. They were selected because they represent the most rigorous and comprehensive available characterizations of the phenomena this document examines. Where those findings are consistent with the MaeV structural interpretation, this consistency is noted. Where they do not address the structural questions the MaeV framework poses, this gap is identified explicitly. Where they raise challenges for the interpretation, those challenges are examined rather than minimized.

The goal of this document is not to marshal evidence for a conclusion already reached. It is to examine, with as much precision as the subject allows, whether a structural pattern is recognizable across independent scientific domains — and what that recognition, if genuine, indicates about the conditions under which meaning-generating systems persist.

Between Stillness and Storm — MaeV Lab, Satellite Document 1

"The question is not what equilibrium looks like from outside. It is what the habitable zone looks like from within — and what it costs to leave it."